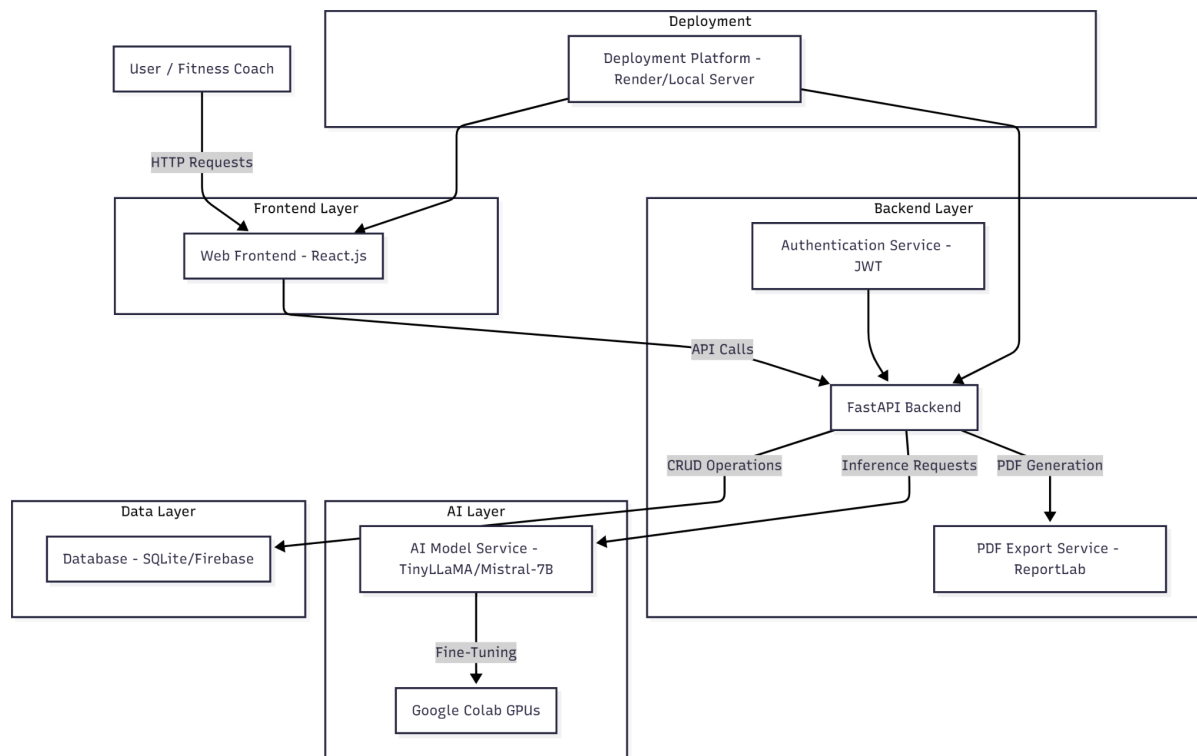




Overall system arch

USER		
int	id	PK
string	username	
string	email	
string	password_hash	
datetime	created_at	



has



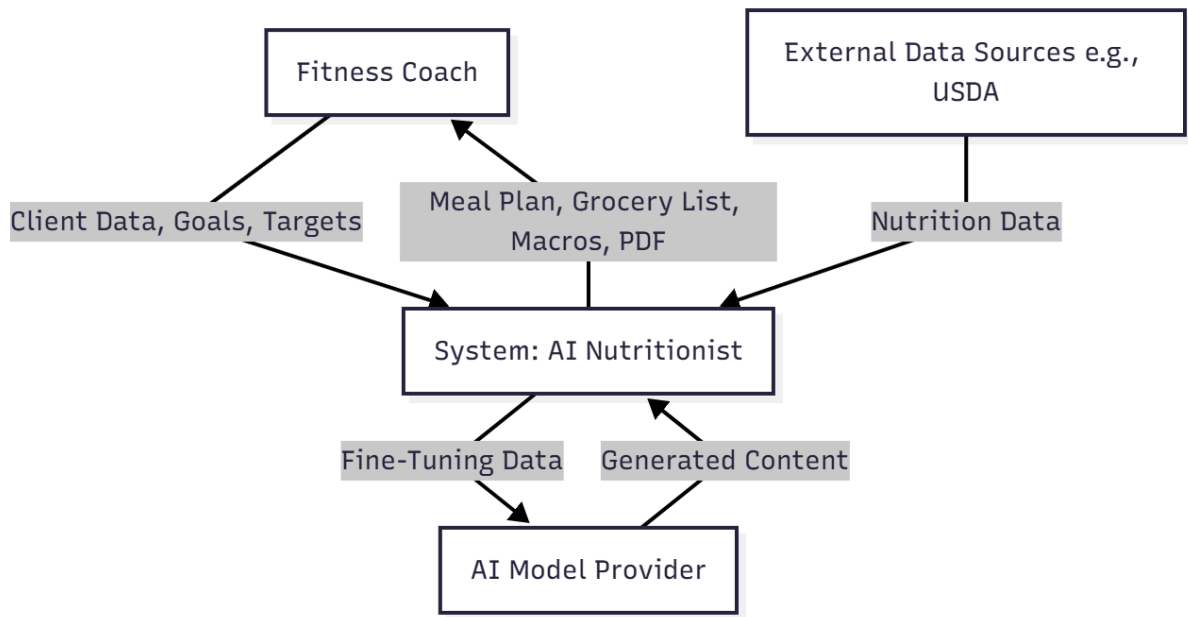
CLIENT			
int	id	PK	
int	user_id	FK	
string	name		
string	goals		
int	calorie_target		
string	diet_type		e.g., vegan, keto
datetime	created_at		



has

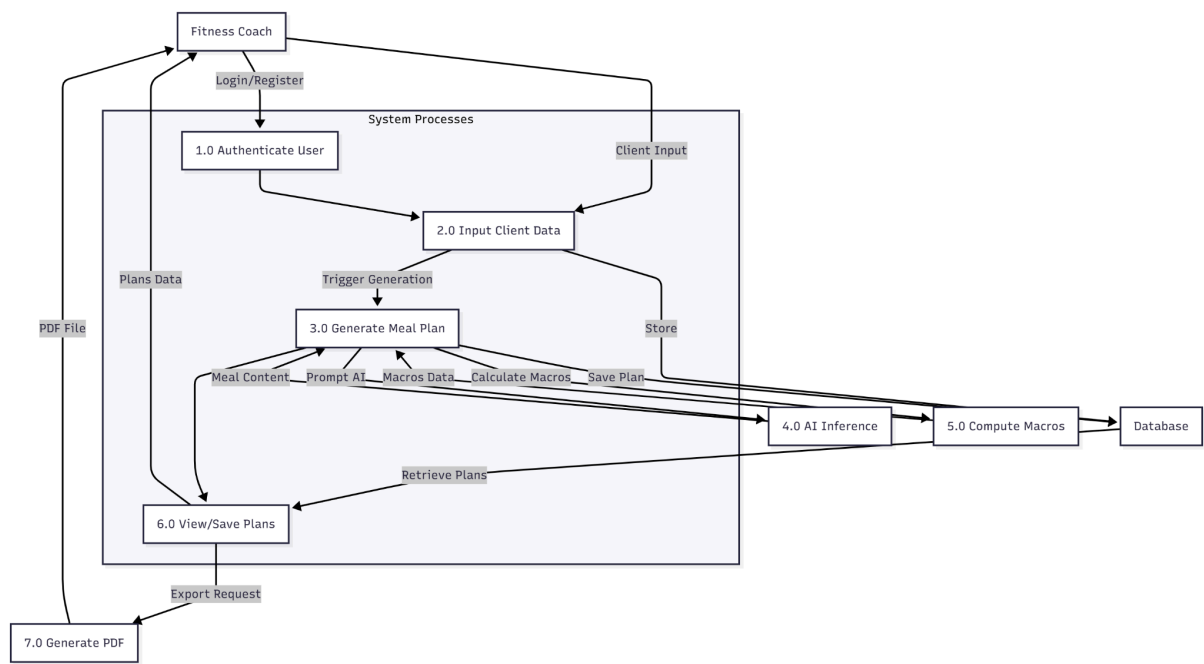


MEAL_PLAN			
int	id	PK	
int	client_id	FK	
datetime	generated_at		
json	plan_details		7-day meal plan JSON
json	grocery_list		JSON array
json	macros		protein, carbs, fats

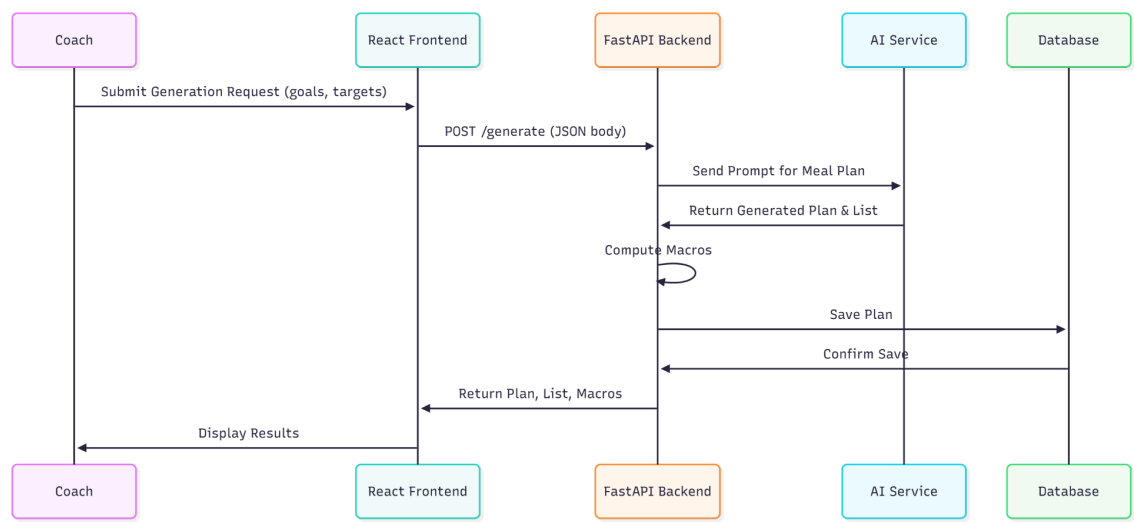


Data Flow diagram level -0

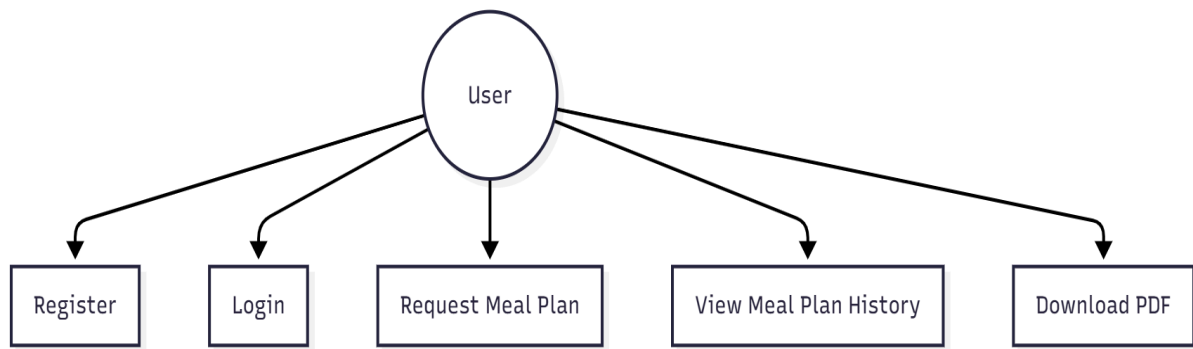
DFD level-1



Sequence diagram



Usecase



Create Detailed API Endpoint Specification

The backend uses RESTful APIs built with FastAPI. Below is a detailed specification for key endpoints, including HTTP methods, paths, parameters, request bodies, responses, and security.

- **/auth/login** (POST)
 - **Description:** Authenticate user and return JWT token.
 - **Request Body:** JSON { "username": string, "password": string }
 - **Responses:**
 - 200 OK: { "access_token": string, "token_type": "bearer" }
 - 401 Unauthorized: { "detail": "Invalid credentials" }
 - **Security:** None (public endpoint).
- **/auth/register** (POST)
 - **Description:** Register a new user.
 - **Request Body:** JSON { "username": string, "email": string, "password": string }
 - **Responses:**
 - 201 Created: { "message": "User created" }
 - 400 Bad Request: { "detail": "Username/email already exists" }
 - **Security:** None.
- **/clients** (POST)
 - **Description:** Create a new client for the authenticated user.
 - **Request Body:** JSON { "name": string, "goals": string, "calorie_target": int, "diet_type": string }
 - **Responses:**
 - 201 Created: { "client_id": int }
 - 400 Bad Request: Validation errors
 - **Security:** JWT required.
- **/generate** (POST)
 - **Description:** Generate meal plan using AI.
 - **Request Body:** JSON { "client_id": int, "goals": string, "calorie_target": int, "diet_type": string }
 - **Responses:**
 - 200 OK: { "meal_plan": json, "grocery_list": json, "macros": json }
 - 500 Internal Error: AI generation failure
 - **Security:** JWT required.
- **/save** (POST)
 - **Description:** Save generated meal plan to DB.
 - **Request Body:** JSON { "client_id": int, "plan_details": json, "grocery_list": json, "macros": json }
 - **Responses:**
 - 201 Created: { "plan_id": int }
 - **Security:** JWT required.
- **/plans/{client_id}** (GET)
 - **Description:** Retrieve saved plans for a client.

- **Path Param:** client_id (int)
- **Query Params:** limit (int, optional, default=10)
- **Responses:**
 - 200 OK: [{ "id": int, "generated_at": datetime, "plan_details": json, ... }]
 - 404 Not Found: Client/plans not found
- **Security:** JWT required.
- **/export/{plan_id}** (GET)
 - **Description:** Export meal plan as PDF.
 - **Path Param:** plan_id (int)
 - **Responses:**
 - 200 OK: Binary PDF file
 - 404 Not Found: Plan not found
 - **Security:** JWT required.

All endpoints include input validation (e.g., via Pydantic), rate limiting, and OWASP protections (e.g., against SQL injection)